Transfer Functions

Embedded Control of Robotics – study notes (Lectures 4 and 5)

Definition

The transfer function of a linear time-invariant (LTI) system is the ratio of the Laplace transform of the output to the Laplace transform of the input, with all initial conditions set to zero.

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G(s) = Y(s) / U(s)
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Y(s) is the Laplace transform of the output, U(s) the Laplace transform of the input and $s = \sigma + j\omega$ the complex frequency variable.

Why it is used

- To check the stability of a system from its poles.
- To find the transient and steady-state response.
- To design controllers and filters.
- To describe a system in the frequency domain.

Deriving a transfer function

1. Write the differential equation of the system.
2. Take the Laplace transform of both sides with zero initial conditions: $d/dt \to s$, $d^2/dt^2 \to s^2$, $\int dt \to 1/s$, $y(t) \to Y(s)$.
3. Factor out Y(s) on one side and U(s) on the other.
4. Divide Y(s) by U(s).

Example 1

```
dy/dt + 3y = u(t)
sY(s) + 3Y(s) = U(s)
Y(s)(s + 3) = U(s)
G(s) = 1 / (s + 3)
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Example 2 (with a gain)

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d²y/dt² + 5 dy/dt + 6y = 2u(t)
(s² + 5s + 6) Y(s) = 2 U(s)
G(s) = 2 / (s² + 5s + 6) = 2 / ((s + 2)(s + 3))
```

Common forms

Form	Expression	Notes
General	N(s) / D(s)	Ratio of two polynomials

Pole–zero	$K(s - z_1)(s - z_2)\ldots / ((s - p_1)(s - p_2)\ldots)$	z = zeros, p = poles
First order	$K / (\tau s + 1)$	One energy-storage element; τ = time constant
Second order	$\omega_n^2 / (s^2 + 2\zeta\omega_n s + \omega_n^2)$	ζ = damping ratio, ω_n = natural frequency

Block diagrams

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Series:    G = G₁ · G₂
Parallel:  G = G₁ + G₂
Negative feedback:  C/R = G / (1 + GH)
Unity negative feedback (H = 1):  C/R = G / (1 + G)
```

First-order system: reading the step response

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G(s) = K / (τs + 1)    step response y(t) = K(1 − e^(−t/τ))
```

- K (DC gain) is the final value of the output for a unit step.
- τ is the time taken to reach 63.2% of the final value.
- Settling time (2%) ≈ 4τ.

Example: G(s) = 4 / (2s + 1) gives K = 4, τ = 2 s, $T_s \approx 8$ s. Written as 2/(s + 0.5), the pole is at −0.5.

Second-order system

Compare the denominator with $s^2 + 2\zeta\omega_n s + \omega_n^2$.

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G(s) = 10 / (s² + 4s + 10)
ωₙ² = 10      →  ωₙ ≈ 3.162 rad/s
2ζωₙ = 4      →  ζ ≈ 0.633
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Damped frequency   ω_d = ωₙ √(1 − ζ²)
Peak time          t_p = π / ω_d
Settling time (2%) Tₛ ≈ 4 / (ζωₙ)
Percent overshoot  PO = 100 · e^(−ζπ / √(1 − ζ²))
```

Example: ζ = 0.5, $\omega_n = 8$ → ω_d = 6.928 rad/s, t_p = 0.454 s, $T_s \approx 1$ s, PO ≈ 16.3%.

Stability from the poles

- Poles −1, −2: both in the left half plane → stable.
- Poles +0.2, −3: one pole in the right half plane → unstable.
- Poles −2 ± j3: left half plane → stable, with oscillation.